Contact Finder & OSINT Engine 🕵️ IN

Comprehensive Project Status & Architecture Report

Repository: `rajakrp18/Project-Researcher`
Primary Model: Gemini 3.6 Flash
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📌 Executive Overview

This monorepo repository (`c:project_researcher`) houses an enterprise-grade Multi-Agent Open Source Intelligence (OSINT) AI engine designed to locate, attribute, and verify Indian mobile numbers based on names, residential/business addresses, companies, and cities. Primary use cases include **IEPF dividend recovery, corporate directory lookups, legal skip tracing, and B2B contact enrichment**.

1. MONOREPO WORKSPACE STRUCTURE

WORKSPACE DIRECTORY	APP NAME	PURPOSE & DESCRIPTION
`apps/contact-finder`	**Contact Finder**	Multi-agent OSINT engine, Puppeteer stealth scrapers, vector embeddings, confidence scoring & Next.js dashboard.
`apps/project-researcher`	**Project Researcher**	Secondary research workspace, PDF parsing tools, and report synthesis AI agent.

2. MULTI-AGENT REASONING ARCHITECTURE

CoordinatorAgent

Master controller. Manages search budgets, batch query lifecycles, corporate entity detection, early exit triggers (≥ 75% confidence), and social proxy searches.

ResearcherAgent

Query generator & fetcher. Builds Google Dorks (combining address, city, pincodes) and coordinates parallel API calls across providers.

ExtractorAgent

3-pass parsing engine: **Pass 1:** Regex matching ('+91'/'0'), **Pass 2:** Semantic vector embeddings, **Pass 3:** LLM context extraction.

VerifierAgent

Identity resolution. Calculates fuzzy string distance (`Fuse.js`), evaluates domain trust, and maintains persistent SQLite database storage.

3. KEY TECHNICAL CAPABILITIES & TECHNOLOGIES

⚡ Upgraded LLM & Vision OCR Core — Gemini 3.6 Flash

Powered by Google's `gemini-3.6-flash` via `@google/genai` (with OpenRouter fallback). If target pages display obfuscated phone numbers as images or HTML canvas elements, **Gemini Vision OCR** captures Base64 screenshots and visually extracts the digits.

🧠 Offline Semantic Vector Embeddings (Transformers.js)

Loads an ONNX-quantized model (`Xenova/all-MiniLM-L6-v2`) locally. Converts webpage paragraphs into vector embeddings and executes Cosine Similarity search completely offline to isolate contact blocks in 30,000+ line documents.

🕵️ Stealth Scraper & Persistent Knowledge Graph

Uses `puppeteer-extra-plugin-stealth` to spoof WebGL fingerprints and bypass Cloudflare/Akamai bot protection. Verifications are saved in a local SQLite (`better-sqlite3`) graph for instant pre-flight cache hits.

4. COMPREHENSIVE CONFIDENCE SCORING & ATTRIBUTION

Total Score (0 – 100%) = `min(100, max(0, S_NameMatch + S_Attribution + S_AddressMatch + S_LocalitySnippet + S_MultiSourceBonus + S_DomainTrust))`

Attribution Classification Matrix

ATTRIBUTION CATEGORY	SCORE IMPACT	DESCRIPTION & ATTRIBUTION CONTEXT
DIRECT_PERSON	+20 PTS	Direct line belonging strictly to the subject or their father (`self` / `father`).
PROXY_PERSON	+10 PTS	Belongs to a colleague, co-director, or associate (`proxy` / `coworker`).
RWA	+5 PTS	Resident Welfare Assoc or Society Secretary (`rwa_official`).
BROKER	+5 PTS	Real estate dealer or property agent handling premises (`broker`).
ORGANIZATION	+3 PTS	Corporate reception desk or general company landline.
GENERIC	0 pts	Standard header lines without specific entity names.
AMBIGUOUS	-5 PTS	Multiple conflicting third-party names listed nearby.
UNKNOWN	-10 PTS	Orphaned phone digits extracted without contextual text.

5. DATA PROVIDER ECOSYSTEM

PROVIDER / SERVICE	STATUS & INTEGRATION DETAILS
SerpAPI (Google)	ACTIVE Key configured in `.env.local`. Primary web search provider.
Gemini 3.6 Flash	ACTIVE Key configured. Drives LLM context parsing and Vision OCR.
Perplexity AI (Sonar Pro)	READY / STANDBY Integrated into `ProviderRegistry.ts` (Priority 15). Pending API key activation.
TinyFish & Explorium	ACTIVE Active keys configured for stealth deep web scraping and B2B lookup.
Directories (ZaubaCorp, IndiaMart, TradeIndia)	ACTIVE Automated scraping modules via stealth Puppeteer.

6. DASHBOARD INTERFACES & SCRIPTS

Interactive Dashboard (`/`)

Target search form with live progress, stop signal, confidence gauge, and interactive evidence trace.

Bulk CSV Processing (`/bulk`)

Processes large datasets concurrently with styled Excel exports (`exceljs`).

Execution Commands

`npm run dev:contact-finder` — Starts Contact Finder dashboard on port 3001.

`npm run dev:researcher` — Starts Project Researcher workspace.